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New MS Concepts for Synthetic Polymers and Additives Characterization

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ABSTRACT

RATIONALE: New ionization processes have been developed for biological mass spectrometry in which the matrix lifts the nonvolatile analyte into the gas phase as ions without any additional energy input. We rationalized that additional fundamental knowledge is needed to assess analytical utility for the field of synthetic polymers and additives.

METHODS: Different mass spectrometers (Thermo Orbitrap (Q-)Exactive (Focus); Waters SYNAPT G2(S)) were employed. The formation of multiply-charged polymer ions upon exposure of the matrix/analyte(/salt) sample to sub-atmospheric pressure directly from the solid state and surfaces facilitates the use of advanced mass spectrometers for detection of polymeric materials including consumer products (e.g., gum).

RESULTS: Astonishingly, using nothing more than a small molecule matrix compound (e.g., 2-methyl-2-nitropropane-1,3-diol or 3-nitrobenzonitrile) and a salt (e.g., mono- or divalent cation(s)), such samples upon exposure to sub-atmospheric pressure transfer nonvolatile polymers *and* nonvolatile salts into the gas phase as multiply-charged ions. These successes contradict the conventional understanding of ionization in MS, because can nonvolatile polymers be lifted in the gas phase as ions not only by as little

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as a volatile matrix but also by the salt required for ionizing the analyte through noncovalent metal cation adduction(s). Prototype *vacuum* MAI and automated sources using a contactless approach are demonstrated for direct analyses of synthetic polymers and plasticizers, minimizing risk of contamination using direct sample introduction to the mass spectrometer vacuum.

CONCLUSIONS: Direct ionization methods from surfaces without the need of high voltage, a laser, or even applied heat are demonstrated for detailed materials characterization using (ultra)high resolution and accurate mass measurements enabled by the multiply-charged ions extending the mass range of high-performance mass spectrometers and use of a split probe. Our vision is that, with further development of fundamentals and dedicated sources, both spatial- and temporal-resolution measurements are within reach if sensitivity is addressed for decreasing sample-size measurements.

INTRODUCTION

Innovative new materials discovery and development to product control and consumer safety are dependent on the aptitude of the analytical resources to characterize them. To achieve best materials performance, physicochemical properties are considered and materials synthesized in which differences are architectural. Spatial and temporal information is needed for characterizing surfaces or reactions from a confined area or time dependent manner. As complexity and structural differences become increasingly important, so does the ability to interrogate these materials more rapidly and from ever decreasing sample sizes. Mass spectrometry (MS) has become an important analytical resource because of its sensitivity, specificity, dynamic range, wide applicability, and high throughput. Since MS interrogates molecular species, it can provide direct molecular composition information from heterogeneous bulk samples and

surfaces, important for educated decisions relative to materials composition.¹ MS has been limited for synthetic polymer characterization especially because of the well-known suppression effects of higher masses in polymers with high polydispersity values (>1.2),^{2,3} relative to other polymer characterization methods, such as gel permeation chromatography, which only offer bulk measurements.⁴ These relatively limited MS developments are in contrast to impressive advances made in the fields of proteomics and metabolomics.⁵⁻⁸ Key to advancing polymeric materials characterization using MS is the need for novel characterization strategies. Some of the more recent improvements in materials characterization are related to ion mobility spectrometry (IMS) MS.^{9,10} Early studies for polymer applications are from the groups of Bowers,^{11,12} de la Mora,¹³ and Clemmer and coworkers^{14,15} using electrospray ionization (ESI) on homebuilt IMS-MS instrumentation. IMS-MS offers an additional separation dimension providing a detailed view of polymeric gas-phase ions including their structures using the information of number of charges and cross-section (size and shape).¹⁶ IMS technology is especially effective when ions are multiply charged.¹⁴⁻¹⁸ Literature shows increasing precedence and maturity for IMS-MS using commercially available instrumentation, permitting inroads into characterization of polymeric materials. This can be largely attributed to the effective deconvolution of polymeric complexity using post-ionization separation of characteristics such as architectural differences.¹⁹⁻²⁴ Improvements and decreasing prices of (ultra)high resolution mass analyzers enabled by Orbitrap technologies have also driven the field, but less so for polymer MS.^{1-6,25-27}

Integral to advances in MS are developments in ionization technology that could provide improvements in simple and direct characterization abilities. Atmospheric pressure and subsequently 'ambient' approaches to ionization have been implemented and a few have shown applicability to synthetic polymer characterization.^{1,28} Limitations of both mass range and IMS separation are expressed in ionization methods that produce only singly-charged ions. Novel ionization processes²⁹⁻³² were recently

discovered producing multiply-charged ions similar to those produced by ESI,³³ but from a solid matrix. These are similar to conditions previously associated with matrix-assisted laser desorption/ionization (MALDI),^{34,35} but without the use of high voltage, laser, heat, or desolvation gases.³⁶⁻³⁹ In matrix-assisted ionization (MAI), the material to be analyzed is placed together with a suitable matrix into the sub-atmospheric pressure region where ionization commences.^{38,39} Multiply-charged ions are produced allowing the use of (ultra)high to low performance mass spectrometers having advanced characterization capabilities including, for example, high mass resolution and mass measurement accuracy,^{40 - 42} IMS gas-phase separations,^{43 , 44} and portability advantages.⁴⁵ MAI-MS analyzed large molecules (66 kDa) by simple exposure of the matrix/analyte to vacuum of a mass spectrometer.³⁹ Interestingly, the matrix/analyte sample can be introduced to the vacuum, available with any atmospheric or intermediate pressure mass spectrometer source, directly from surfaces with no or minor instrument modifications.^{39,41,43,46-48}

Currently, the most analytically useful matrix, 3-nitrobenzonitrile (3-NBN), preferentially ionizes more basic compounds and materials containing nitrogen(s) including drugs, peptides, and proteins by the addition or removal of one or more protons.^{38,42,49-51} The addition of a variety of ammonium salts has been shown helpful for analytes that prefer protonation⁵² because these conditions effectively suppress ionization of the synthetic polymers which may be present in some samples such as peptides. Therefore the applicability of ionizing synthetic polymers by MAI-MS has been limited because of the requirement of metal cation attachment and the inefficiency of doing so using 3-NBN matrix at the typical relative low inlet temperatures.^{43,44} However, when the matrix is simply the solvent, as in solvent assisted ionization (SAI),³⁶ used to dissolve the polymer, metal cation attachment is achievable when a heated inlet tube (Thermo IonMax source) is used at high temperatures (~400 °C). Consequently, a compound screening study, with the objective to perform MAI more efficiently, broadly,

or specifically, discovered over 40 ionizing matrices operational at room, slightly elevated, or freezing temperatures (<0 to ~80 °C) of an intermediate pressure MALDI source or a Z-Spray ESI source (Waters);⁵³ developments in analogy to matrix studies after MALDI and ESI (note, the solvent is also a matrix) were invented.⁵⁴⁻⁵⁶ One of the new MAI matrices, 1,2-dicyanobenzene (1,2-DCB), efficiently ionized, by protonation, complex structures containing europium, poly(ethylene glycol) (PEG), and up to a dozen amines and amides when the matrix/analyte sample was directly introduced from atmospheric pressure.⁵⁷ An interesting trend has been observed for synthetic polymers using various matrices from atmospheric and intermediate pressure: improved ion abundance and higher charge states are achieved from vacuum conditions. This is contrary to the analyses of small proteins where higher pressure provided higher charge states.⁵³

We speculated that the (in)ability to ionize synthetic polymers by metal cation adduction may be related to the (im)proper *inlet* and *vacuum* ionization conditions, especially in the absence of added energy (heat) provided for the ionization to proceed. Additional factors are the obviously important sample preparation conditions of matrix and salt among other factors, well described in the literature for polymer MS analyses in general.^{2,3,58-61} Furthermore, despite several detailed studies, the use of salt additives may not have been fully explored for applications in synthetic polymer MS more generally.⁶²⁻⁶⁸ We describe the development of new ionization processes²⁸⁻³² as efficient and robust ionization technologies and methods for additives and low polydispersity synthetic polymers (<10 kDa). The potential utility for *direct* materials characterization from a variety of surfaces and in a timely manner using commercial high performance mass spectrometers, including IMS-MS instruments, with little to no source modification and, most effectively, dedicated sources⁶⁹⁻⁷² based on a contactless and automated introduction directly from atmospheric pressure to the sub-atmospheric pressure are demonstrated and fundamentals discussed.

EXPERIMENTAL

Materials: Matrix compounds including 3-NBN, methyl 2-methyl-3-nitrobenzoate, bronopol, dithranol, 3-nitrobenzoyl chloride, as well as salts were purchased from Sigma-Aldrich (St. Louis, MO, USA) unless otherwise noted, as well as HPLC methanol and water (H₂O), fexofenadine HCl, bradykinin, leucine enkephalin, methionine enkephalin, PEG ethyl hexanoate 640 (PEG-EH), 6-(dibutylamino)-1,8-diazabicyclo[5.4.0]undec-7-ene (DADBU), PEG dimethyl ether-1500 (PEGDME), and Bisphenol-A. Formic acid was obtained through EMD Millipore (Burlington, MA, USA) and acetonitrile (ACN) was from Fisher (Pittsburgh, PA, USA). PEG-1000, 3-methyl-4-nitroaniline, 2-methyl-2-nitropropan-1,3-diol and 1,2-DCB were purchased from Alfa Aesar (Ward Hill, MA, USA). PEG-1500 and PEG-3000 were purchased from PSS Polymers. The matrix α -cyano-4-hydroxycinnamic acid (CHCA) was from Waters Corp. (Milford, MA, USA).

Automated and Contactless MAI Directly from Atmospheric Pressure: An automated XYZ-stage, previously employed for MS imaging applications,⁷³ was programmed to align the sample/matrix spot with the MSTM heated inlet tube⁷⁴ that was retrofitted to replace the skimmer cone assembly on the ion block of a SYNAPT G2S mass spectrometer (Waters Corp.). The arm holding the plate containing the spots of matrix/analyte/salt sample is positioned on a 'log spot' from which the automated XYZ-movement samples the different contents sequentially by moving each spot in a continuous fashion in front of the mass spectrometer inlet at a distance of ~1 mm to allow the individual spot to be 'vacuumed' into the inlet without physical contact. This contactless removal from the surface into the inlet of the mass spectrometer is associated with the pressure differential inherent with the instrument operation^{38,75} and the heat applied to the retrofitted MSTM inlet tube.^{71,74} This alignment remains for

typically 1 second per sample and the automated setup then positions the next spot by withdrawing away in the Z direction and moving on to the next spot in the X direction. Upon positioning, the stage again moves towards the mass spectrometer inlet in the Z direction at ~1 mm distance relative to the inlet of the mass spectrometer. A movie of the principal process was published elsewhere.⁷¹ The matrix/analyte/salt spots were prepared on a clean glass slide and, to guide the eye, using a commercial steel MALDI plate in the background as reference for the distance between sample spots. The spotting was performed in layered fashion by placing 0.5 μL of sample and then carefully adding 0.5 μL of matrix on top of it. The polymers and plasticizers were prepared in a solution of ACN:water 1:1 (v/v) and the matrix (3-NBN, 0.1 mg μL^{-1}) was prepared in 100% ACN.

Accurate and Fast Mass Measurements using the Vacuum-Probe vMAI Sources: A prototype intermediate pressure vMAI source⁶⁹ was fitted to a Waters SYNAPT G2S IMS-TOF mass spectrometer. The split probe⁷¹ was used here with two consecutive probe introductions. The first probe introduction contained only an external peptide standard with matrix. The external peptide standard mixture contained 2 ng μL^{-1} each of leucine enkephalin and methionine enkephalin. Onto one surface of a split-tip vMAI probe was pipetted 1 μL of the external standard mixture along with 1 μL 3-NBN solution (0.1 mg μL^{-1} in ACN) and briefly dried in air. The probe was then inserted into the vacuum source, and remained until the peptide ion abundance reduced to approximately 1.0×10^3 counts per second, at which point the probe was removed. The second probe introduction contained not only the remaining peptide/matrix mixture, but also the matrix/analyte/salt combination on the other surface. PEG-1000 at a concentration of 20 pmol μL^{-1} in 1:1 ACN:H₂O, 1,2-DCB matrix (5 mg in 50 μL 3:1 ACN:H₂O), and aqueous sodium trifluoroacetate (1 mg mL^{-1}) were mixed in equal volumes, and 3 μL of the resulting solution was pipetted onto the second surface of the split-tip vMAI probe. After drying in air, the probe now containing both types of

matrix/analyte(/salt) samples, physically separated, was re-inserted into the source. The resulting mass spectrum contained both the external standards and the synthetic polymer distribution in good ion abundance. The commonly used external peptide standards leucine enkephalin ($[M+H]^+$ with mass-to-charge (m/z) 556.2771) and methionine enkephalin ($[M+H]^+$ with m/z 574.2337) were used to determine the cation attached to the polymer backbone using the accurate mass of a doubly-charged PEG-1000 ion that is close in m/z value, in this case the nominal mass of m/z 560. The resulting mass spectrum of the simultaneous ionization of both PEG-1000 and the external peptide standard mixture from the two different sides of the split probe was centroided, and the singly-charged ion of leucine enkephalin signal (m/z 556.2771), selected because of its closeness to the doubly-charged PEG ion with the nominal mass of m/z 560, was corrected to give the accurate mass through a single-point calibration.

The prototype *vacuum* source on a Q-Exactive Focus hybrid quadrupole Orbitrap mass spectrometer (Thermo Fisher Scientific, Waltham, MA, USA)⁷⁰ was used for the accurate mass measurements of plasticizers. Bisphenol A (20 pmol μL^{-1}) was prepared in 50 mM ammonium acetate in a ratio of 1:1 (v/v) ACN:water and 3-NBN matrix solutions were prepared in 3:1 (v/v) ACN:water. The premixed solution of analyte and matrix (3-NBN) in a ratio of 1:1 (v/v) was spotted onto the tip of probe and was air dried before insertion.

vMAI-IMS-MS of polymer-additive mixtures using the prototype vMAI source fitted onto a Waters SYNAPT G2S IMS-TOF mass spectrometer⁶⁹ was based on low molecular weight PEG with different end-groups, a plasticizer, and a property modifier, similar to previous tertiary blend analyses.⁴⁴ PEG-1000 was prepared to a concentration of 20 pmol μL^{-1} in 1:1 ACN:H₂O, the plasticizer PEG-EH to 10 pmol μL^{-1} in 1:1 ACN:H₂O, and DADBU to a concentration of 1 pmol μL^{-1} in 1:1 ACN:H₂O. In addition to 3 μL of

aqueous sodium trifluoroacetate (1 mg mL^{-1}) and $3 \text{ }\mu\text{L}$ 3-NBN (0.1 mg mL^{-1} in ACN), $1 \text{ }\mu\text{L}$ aliquots of each of the three analytes were vortexed together, and $3 \text{ }\mu\text{L}$ of this solution was spotted on the end of a probe for use with the prototype vMAI source⁶⁹ fitted onto the SYNAPT G2S IMS-TOF mass spectrometer.

Comparison of Salts for MAI: PEG-1000 ($20 \text{ pmol }\mu\text{L}^{-1}$ in 1:1 ACN:H₂O) was mixed in equal portions with an aqueous salt solution (1 mg mL^{-1} barium chloride, strontium chloride, lithium chloride, or sodium trifluoroacetate) as well as matrix (3-NBN, 0.1 mg mL^{-1} in ACN; methyl 2-methyl-3-nitrobenzoate, 0.1 mg mL^{-1} in 3:1 ACN:H₂O; or 1,2-DCB, 0.1 mg mL^{-1} in 3:1 ACN:H₂O). Alternatively, the analyte, salt, and matrix may simply be pipetted onto the probe tip without being previously mixed. From the resulting mixture an aliquot of $3 \text{ }\mu\text{L}$ was spotted and dried on the end of the vMAI probe, which was inserted into the prototype vMAI source⁶⁹ on a SYNAPT G2S IMS-TOF mass spectrometer.

Accurate Mass Measurements of PEG-1000 using MALDI: An external standard consisting of $1 \text{ }\mu\text{L}$ bradykinin solution ($20 \text{ pmol }\mu\text{L}^{-1}$ in 1:1 ACN:H₂O) mixed with $1 \text{ }\mu\text{L}$ CHCA (10 mg mL^{-1} in 1:1 ACN:H₂O) matrix was spotted onto a commercial MALDI plate using the dried droplet method. Next, $1 \text{ }\mu\text{L}$ PEG-1000 ($20 \text{ pmol }\mu\text{L}^{-1}$ in 1:1 ACN:H₂O) mixed in equal parts with aqueous sodium trifluoroacetate (1 mg mL^{-1}) and dithranol (10 mg mL^{-1} in tetrahydrofuran) was also spotted on an adjacent sample spot. This target plate was loaded into a SYNAPT G2S IMS-TOF mass spectrometer, and both matrix/analyte(/salt) spots were consecutively desorbed, each for 20 seconds, with the laser power at 200 (arbitrary units) and a firing rate of 1000 Hz. The mass spectra were summed and centroided. The $[\text{M}+\text{H}]^+$ bradykinin ion, corrected to reflect a mass of m/z 1060.5692, was used for single-point calibration. The nearest PEG signal was located at a nominal mass of m/z 1053 and used for the accurate mass measurement with MALDI.

RESULTS AND DISCUSSION

This work is the first paper dedicated to the application of polymers and additives using the new ionization processes, other than a proof-of-principle study,⁴⁴ with special emphasis on (v)MAI from atmospheric pressure (*inlet* ionization) and vacuum (*vacuum* ionization) because of its uniqueness, simplicity, analytical utility, robustness, and compatibility with specific source geometries. The following section addresses the fundamental aspects clarifying that ionization proceeds through metal cation attachment, the implementation on commercial sources with minor modifications, and the use of dedicated platforms and sources to determine the new ionization processes more fully for low-polydispersity (<1.2) low-molecular weight (<10 kDa) polymer and additive applications.

Fundamental Aspects of Polymer Ionization using New Ionization Processes

Using the simple pipet approach⁴¹ various additives were ionized and detected using the 3-NBN matrix (**Figure S1, supporting information**) on the overridden ESI source of the SYNAPT G2 mass spectrometer. Low molecular weight polymers using the MAI matrix 3-NBN ionized poorly, if at all, using the atmospheric pressure inlet of the SYNAPT G2, in contrast to peptides and other biological materials which ionize readily by protonation,^{38,39} and in fact discriminate against metal cation adduction using the new ionization processes.⁴³ Further work performed on testing conditions for low molecular weight polymer analyses, both the more obvious conditions considered to be important for the MS analysis of synthetic polymers and conditions that have been shown to be important or to allow the manipulation of the new ionization processes were applied.⁴⁴ The six MAI matrices tested (**Figure S2, supporting information**), provided typical doubly-charged ESI-like ions of PEG-1000 using the typical low temperatures of a heated inlet of the commercial skimmer cone Z-Spray source. Based on the analyte ion intensities and the charge states observed, the relative order of performance was 3-NBN ~ 3-nitrobenzoylchloride < 3-methyl-4-nitroaniline ~ 1,3-dinitrobenzene < bronopol

< 2-methyl-2-nitro-1,3-propanediol. The effect of the Z-Spray sampling cone inlet temperature was studied by increasing the temperature of the source block from 50 to 150 °C (maximum temperature commercially available by this vendor) using the matrix bronopol. The best results were acquired at 50 °C, and the analyte ion abundance completely ceased at 150 °C (**Figure S3, supporting information**). Using three different salts (LiTFA, NaTFA, KTFA), typically added in polymer characterization using MS,^{61,76} suggests that the added cation probably attaches to the polymer backbone (**Figure S4, supporting information**). The addition of NaTFA provided the most and KTFA the least abundant polymer ions using the bronopol matrix. In the case of NaTFA even what appear to be triply-sodiated ions, were detected with the most abundant ions being doubly charged. Similarly, the matrix 2-methyl-2-nitro-1,3-propanediol with the addition of NaTFA gives good analyte ion abundance of the doubly-charged polymer ions (**Figure S5, supporting information**).

The ionization processes that underlie these simple approaches are not easily understood. Detailed accounts on fundamentals using biological polymers can be found in refs. [29-32, 53]. However, the results presented here are noticeably more complex because of the necessity for metal cation attachment, requiring explanations relative to the new ionization processes and comparison with traditional ionization methods such as ESI and MALDI. The importance of adding salt additives to ionize polymers is well known,^{58-60,61,76} but metal cation attachment has been associated in MALDI with gas-phase reactions of bare analyte molecules with metal cations present, believed to be initiated by a laser.⁷⁷ It is difficult to conceive how such a high-energy process could occur under these low temperature MAI conditions without the employment of voltages, high heat, laser, or even desolvation gases. To be clear, using basic fundamental knowledge known and applied for decades for use in MS, the described ionization of polymers should not be possible. The MAI process has been hypothesized to be a sublimation-driven triboluminescence-like process.^{30,31} It is, in particular, not easily

understood how salts can be associated with sublimation-driven processes.⁷⁸⁻⁸⁰ It is surprising and fundamentally complicated that salty conditions lead to charge separation to form the gaseous charged clusters from which the matrix and residual salt would need to sublime/evaporate to unravel the naked polymer-metal cation associations. However, we have no reason to believe otherwise than that the processes are fundamentally different between the synthetic and biological polymer ionization other than their basicities and therefore preferences for attaching metal cation(s) *versus* proton(s) as is the case with any ionization method.⁷⁷ We also do not believe that there is a fundamental difference between *inlet* and *vacuum* ionization in which one uses, simply put, an inlet tube and in another the matrix/analyte(/salt) samples are placed directly into the vacuum of the mass spectrometer.

This astonishing ability to ‘simply’ sublime a volatile matrix together with a nonvolatile analyte *and* a salt to create abundant, gaseous ESI-like charge states of $[\text{polymer}+n(\text{metal cation})]^{n+}$ ions only by the exposure to sub-atmospheric pressure is shown in **Figure 1**. *Vacuum* MAI was implemented on an intermediate pressure MALDI source of a SYNAPT G2S and G2 without the use of a laser employing the MAI matrix 3-NBN. Using the extraction voltage and hexapole settings typically used for MALDI operation, the nonvolatile PEG-DME doped with LiCl and cocrystallized with 3-NBN matrix (**Figure 1A**) provides the typical 2-dimensional output (**Figure 1D**) of sigmoidal transitions (**Figure 1C**), as previously reported for ESI-IMS-MS.¹⁵ The ionization lasted about 16 minutes and proceeded in a continuous fashion (**Figure 1B**), as is the case in ESI, but from the solid state at intermediate pressure without using a laser or high voltages. Unlike ESI, which is prone to inlet and source contamination, vMAI, performed on an intermediate pressure MALDI source without engaging the laser, is exceptionally robust. While the processes are not well understood. By simply adding heat on a heated inlet tube, it is bewildering that, by simply exposing the matrix/analyte/salt to sub-atmospheric pressure, nonvolatile analytes *and* salts are

taken into the gas phase to form gaseous and multiply-charged ions, essentially just like ESI (**Figure S6, supporting information**). Therefore polymer ionization can quite effectively proceed directly from sub-atmospheric pressure employing the new ionization processes.⁴⁴ The analytical utility of performing vMAI mass measurements utilizing a commercial intermediate pressure MALDI source but not using the expensive laser, is obviously not economically viable. In addition, it often requires several minutes to introduce a sample into a commercial vacuum MALDI source because of the necessity to evacuate the source after the plate was loaded. Quicker and less expensive solutions are presented below.

An interesting unresolved finding is not only the unique observation of ‘watching’ polymers and salts being transferred into the gas phase by witnessing the ‘disappearance’ of the matrix:polymer:salt sample on an intermediate pressure source without the use of a laser, as was captured here by screenshots of the display using the commercial embedded camera, through the ‘simple’ assistance of the matrix, bronopol in this specific case (**Figure 2**; and **Figure S7 and S8, supporting information**), but also the detection of doubly-charged polymer ions that mathematically calculate to +40 for the sum of the two cation masses. This therefore may suggest a combination of H^+ and K^+ for analyte ionization (while this potential explanation is unconfirmed, we have observed the similar mass difference in other measurements including proteins when present in low abundance, 50 fmol).⁸¹ The +40 polymer distribution observed here ceases from the matrix/analyte(/salt) sample surface prior to that of the potentially doubly-sodiated species, $[M+2Na]^{2+}$. This is evidenced in the mass spectra extracted from the total ion chromatogram (TIC) (**Figure S7, supporting information**). These mass spectral results obtained from the TIC indicate that the specific +40 polymer ions may be more volatile and/or surface active than $[M+2Na]^{2+}$. According to the mass spectra (**Figure S8, supporting information**) extracted from the 2-dimensional IMS-MS plot (**Figure 2**), the +40 ions (charge state family with purple line to guide the eye) show

faster drift times than the $[M+2Na]^{2+}$ ions (charge state family with red line to guide the eye), indicating more compact gaseous ion structures. While there is a significant amount of information extractable through MS, especially in combination with IMS, other technologies such as spectroscopy may be more likely to decipher the fundamentals of these new processes in the future.

The fact that heat can be used to speed up the ionization process is consistent with a phase diagram associated with an evaporation (instead of sublimation) of a compound (matrix) at increased temperatures (note a phase diagram does not provide any information relative to charge separation). We constructed a simple device that allows the use of both temperature and sub-atmospheric pressure to be applied to the mass measurements (**Scheme 1**; and **Scheme S1, supporting information**). The following movie exemplifies the process and the simplicity of the measurements using MAI for the analyses of polymers (**Movie S1, supporting information**), providing a general feel of temperature and pressure conditions having a synergetic relationship. Comparisons using this simple inlet modification are shown in **Figure S9 (supporting information)**, for 3-NBN and 2-methyl-2-nitro-1,3-propanediol matrices ionizing PEG-3000 in the presence of the salt $Ba(OAc)_2$. While the matrix 3-NBN exclusively generates quadruply-charged PEG-3000 ions in this matrix/analyte/salt mixture, the abundance of this distribution is only 2.05×10^3 counts. When this matrix is replaced by 2-methyl-2-nitro-1,3-propanediol, doubly-, triply-, and quadruply-charged ions are observed with an abundance of 1.47×10^4 counts. The latter matrix therefore provides both alternative charge state distributions and increased analyte ion abundance.

Importantly these initial results exemplify how a polymer sample containing salt and matrix enables the formation of nonvolatile, gaseous metal cation-adducted polymeric ions from conditions that do not appear amenable based on traditional MS knowledge. These results indicate that the key to improving polymer ionization at, or near, atmospheric pressure involves several factors: the pressure (pressure differential) itself,

the heat, and importantly, the choice of matrix. These conclusions are similar to those of previous studies in which the highest molecular weight analyte (bovine serum albumin, 66 kDa) could be analyzed by the MAI process with the matrices 3-NBN or 2-nitrophenylglucitol, using either a heated inlet tube directly from atmospheric pressure or a vacuum source without added heat.^{31,39,82,83} In some of these examples, thermal energy was added by either a laser or a heated inlet tube. With the polymer samples it is apparent that the exposure directly to sub-atmospheric pressure combined with an appropriate matrix is a key factor for improving the analytical utility of the new ionization processes for these salty types of samples. This does not imply any specific conditions relative to *vacuum* ionization, but one can expect that increased heat will be helpful as is the case with *inlet* ionization and vice versa.

It has been well documented that ionization via metal cation attachment to biological polymers (e.g., peptides and proteins) is significantly suppressed using the 3-NBN matrix and it is therefore considered to be a well performing matrix for biological materials in positive ion mode.^{30,31,38,39,41-43,75,84} While it is a highly desirable feature to avoid metal cation attachment to the analyte for biological MS, it is typically a necessity for synthetic polymer MS.^{1-3,5,6,58-61,63,65,68,76,77} To investigate metal cationization, accurate *v*MAI mass measurements of the gas-phase ions were performed using the prototype *vacuum* ionization source⁶⁹ on the SYNAPT G2S (note, the use of the dedicated *vacuum* ionization source as an analytical tool is described in the last section of this work). The mass measurements confirm that the ions under these specific conditions are typical ESI-like ions (**Figure S10, supporting information**). To establish the principal means of ionization using MAI-MS, accurate mass measurements were performed using PEG-1000. PEG, doubly-charged with two sodium cation adductions, $[\text{RO}(\text{CH}_2\text{CH}_2\text{O})_n\text{R}' + 2\text{Na}]^{2+}$, has the same nominal mass as the doubly-protonated polymer with one extra repeating unit, $[\text{RO}(\text{CH}_2\text{CH}_2\text{O})_{n+1}\text{R}' + 2\text{H}]^{2+}$. Using the vacuum probe⁶⁹ with a split-probe tip,⁷¹ the analyte

and standards were ionized simultaneously from the respective split-probe sides after the ion abundance of the easier to ionize peptide was briefly monitored to give an ion abundance of about high e^3 after which the introduction of both the analyte to be mass calibrated (polymer) and the external standard was performed. Without two consecutive introductions, the abundance of the polymer ions was too low in comparison with that of the highly-abundant peptide ions. The resulting mass spectrum showed the concurrent ionization of PEG-1000 and both peptide standards. The leucine enkephalin signal at m/z 556 was close to a doubly-charged PEG ion occurring at m/z 560, and therefore this peptide was chosen for single-point calibration of the centroided isotopically resolved signal. From this, accurate mass measurements of the doubly-charged PEG ion were performed at m/z 560.3090. The $[\text{HO}(\text{CH}_2\text{CH}_2\text{O})_{24}\text{H}+2\text{Na}]^{2+}$ ion was computed to be m/z 560.3097 while the $[\text{HO}(\text{CH}_2\text{CH}_2\text{O})_{25}\text{H}+2\text{H}]^{2+}$ ion was calculated at m/z 560.3408. The experimentally determined mass values differ by 0.7 mDa and 31.1 mDa, respectively, strongly suggesting that the PEG forms doubly-sodiated adducts rather than protonated species by the new ionization process (vMAI) without added energy input. Metal cation attachment to the polymer chain is in good agreement with the ESI-MS of PEGs.^{13-22,24-26} This principal mass calibration approach was also tested by directly spotting the respective samples on the same probe tip or, alternatively, spotting the different samples onto the different sides of the split probe but with immediate introduction of both samples at the same time. There are undesired effects during sample crystallization in the former and significantly different ease in ion formation in the latter so that both these approaches failed to provide useful ion abundance. The doubly-charged PEG ion with a nominal value of m/z 560 provided a reliable means for accurate mass measurements with a single point calibration because of the robust polymer ion abundance and the closeness of the m/z value to those of the external peptide standards using the split probe. The ability to submit differently ionizing compounds and from the solid state for accurate mass measurements and

calibration is also expected to be of value for imaging and spatial resolution applications of materials such as films.

The accurate mass measurements of the same PEG-1000 were also made using traditional MALDI-MS on the commercial intermediate pressure MALDI source of the same SYNAPT G2S mass spectrometer with the commercial Nd:YAG laser (355 nm) at relatively low laser fluence. The instrument was first calibrated using a mixture of the matrix CHCA and the peptide bradykinin as an external standard. The mass spectrum of the following spot containing the polymer PEG-1000 with the dithranol matrix and sodium trifluoroacetate salt showed exclusively singly-charged sodium adducts under these conditions. Interestingly, the mass accuracy of this MALDI-MS experiment was calculated to be 1.6 mDa off (**Figure S11, supporting information**), a three-fold increase relative to vMAI of the same PEG-1000 sample on the same instrument. With MALDI-TOF, one can also expect ion dispersion effects related to angular and energy dispersion, but with Q-TOF neither should matter. We hypothesize that the improved results using vMAI relate to the multiply-charged ions bringing the m/z values of the ions in a range of improved mass resolution and accuracy. Previous reports have shown improvements in mass resolution using MAI on a QTOF to be about 100,000 when the instrument using ESI is benchmarked at ca 60,000.⁵³

Caution must be given to potential source contamination with any direct injection methods to a commercial ESI source to perform either ESI or MAI, especially when salts are required to also be added because there are intrinsically physical contacts between sample and atmospheric pressure ionization mass spectrometer inlets. On the contrary, using the commercial vacuum MALDI source these specific limitations do not apply, especially when no laser (vMAI) is used to lift the matrix/analyte(/salt) sample into the gas phase where ionization proceeds. Instead, the time that it takes to introduce the matrix/analyte(/salt) to vacuum is a specific current limitation using the commercial MALDI source for vMAI (no laser), in addition to the financial burden associated with

purchasing a MALDI source. In the following sections we first address the method development and use the knowledge for dedicated source applications for synthetic materials.

Method Development using Commercial Mass Spectrometer Inlets

Obviously, many factors can be expected to have an influence on the MS outcome as is the case with traditional ionization methods. First, we evaluated the outcome of how a sample is presented to the inlet of a mass spectrometer source. Performing the same experiment from a glass plate using the matrix/analyte/salt from a flat surface relative to that of a pipet tip introduction exemplifies a more daunting task: to lift the sample into the gas phase from the glass plate even from the similar distance to the inlet, while the matrix/analyte(/salt) on a pointy pipet tip is significantly easier to draw into the vacuum of the mass spectrometer. Importantly, the ion abundance of the polymer ions was notably better with conditions that made use of the vacuum *versus* the rapid matrix/analyte/salt introduction using a pipet tip. These results suggest that the charging/declustering of these salty samples could not occur in the short time scale especially associated with the simple pipet tip approach, contrary to that of the biological polymers.^{75,84} Further improvements for polymer ionization can be achieved by ‘sealing off’ the mass spectrometer inlet by a glass plate, resulting in sub-atmospheric introduction conditions similar to previous work with tissue,³⁹ while preferentially implementing a gas leak to still provide some airflow for ‘healthy’ mass spectrometer operation. A gas leak is also accommodated with the commercial vacuum MALDI source on the same mass spectrometer. The spontaneous nature of MAI for ionizing synthetic polymers directly from a surface operated from atmospheric pressure has been noted above with **Movie S1 (supporting information)**. Alternatively, simply placing the matrix/analyte/salt sample inside the skimmer cone of the commercial ESI source before opening the isolation valve allows direct exposure of the matrix/analyte(/salt) sample to sub-atmospheric pressure, aiding in good MAI ion

abundance of the polymer. Obviously, none of these test approaches provide analytically useful methods but rather guide in what is helpful for a method useful for synthetic polymer analyses.

A MAI temperature study using the simple modification described in **Scheme S1 (supporting information)** on the commercial Z-Spray ESI source of the SYNAPT G2 mass spectrometer suggests that under these specific conditions (**Figure 3A**), the ionization prefers low to no temperature applied to the source block. The effects of cations (**Figure 3B**) and counter anions (**Figure 3C**) were also tested in more detail on PEG-1000 and PEG-3000 using 2-methyl-2-nitropropan-1,3-diol as a matrix on the SYNAPT G2 with the source block set at 55 °C. The highest ion abundance for PEG-1000 (**Figure 3(1)**) was obtained for Ba(OAc)₂ whereas for PEG-3000 (**Figure 3(2)**) the difference between the different divalent salts is less obvious. Under the experimental conditions used here, changing the counter anion of the salt shows only a limited impact on the ionization success of both PEG sizes (**Figure 3C**). These results show that the cations, added in the form of metal salts, can strongly guide the ability of ionizing synthetic polymers by MAI, as is the case with traditional ionization methods.^{11-22,24-26}

A comparison between vMAI, ESI, and vacuum MALDI on the SYNAPT G2 (**Figure S12, supporting information**) shows a trend that the new ionization processes provide charge states that are nearly as high as ESI under the conditions used. A comparison of the drift-time distribution is shown in **Figure S12 (supporting information)** exemplifying how the vMAI approach provides excellent IMS separation similar to ESI whereas MALDI and atmospheric pressure chemical ionization (APCI)-type ionization approaches resulting in singly-charged ions do not. vMAI-IMS-MS makes more effective use of the 2-dimensional (2-D) space (peak capacity) of drift time and *m/z* separation. Changes in charge states have been previously shown to be associated with higher energy conditions required to form the lower charge states.²⁹ The drift times are essentially identical for the multiply-charged ions produced by vMAI and ESI

(**Figure S13A, supporting information**). Drift-time comparisons between divalent and monovalent cations (Ba^{2+} , Ca^{2+} , Sr^{2+} , and Na^{+}) show that there is a cation size dependence and that the best separation of closely related m/z values is typically more readily achieved with increasing charge states and/or cationic radii (**Figure S13B, supporting information**), as one must expect from previous work using ESI-IMS-MS.¹³⁻²¹

PEG-10000 was ionized using 2-methyl-2-nitropropan-1,3-diol matrix and calcium acetate salt in a 1:100 molar ratio (**Figure 4**). The data acquisitions, typically a couple minutes, included the use of IMS in addition to MS, to decipher the different charge states by extraction from the 2-D plot of drift time and m/z separations (**Figure S14, supporting information**). Using the rapid vMAI-IMS-MS approach, the observed 'pictorial snapshots', as previously proposed for ESI-IMS-MS,¹⁵ appear to be distinctive enough. As one specific example, consumer products (**Figure S15, supporting information**) such as lens cleaner (**Figure S16, supporting information**), adhesive tape (**Figure S17, supporting information**), and chewing gum (**Figure S18, supporting information**) were readily ionized directly from their own surfaces. The potential m/z assignments for the chewing gum are according to those by Wilkins and coworkers using a MALDI-TOF mass spectrometer.⁸⁵ The analytical applicability for these low-mass polymers suggests that vMAI is an alternative method to MALDI, but does not require a laser, making the experiments less expensive and simpler, while extracting at least the same and typically more information because of the extended mass range and improved IMS separation. The utility of the formation of multiply-charged ions in an experiment that looks like MALDI but does not employ a laser is attractive for coupling to high performance mass spectrometers because of the IMS and/or accurate, high resolution mass measurements, in addition to being more robust, and eliminating source contamination.

The analyses of materials composition directly from surfaces has significant potential utility. Desirable features are to have the capability to analyze multiple samples and in a short period of time, similar to MALDI, and preferably operated from atmospheric pressure, similar to ESI. A recently established, dedicated automated ionization platform was used in the following (**Figure 5A**; and **Figures S19A** and **S20A, supporting information**); a movie of the operation was presented in the initial publication.⁷¹ This contactless, automated operation directly from atmospheric pressure is based on desirable features using the concept of *inlet* ionization in which the mass spectrometer inlet tube retrofitted to the Z-Spray ion block uses both heat and sub-atmospheric pressure, to draw the matrix/analyte/salt sample from the approaching surface into the sub-atmospheric pressure region of the SYNAPT G2S mass spectrometer, acting similarly to a 'vacuum cleaner' without physical contact between the surface and the inlet. The area of matrix/analyte(/salt) which is consumed from the surface is of similar size to the 500 μm inner diameter inlet tube. Automated results were readily obtained from atmospheric pressure using the XYZ-stage movement in all three directions to address cross-contamination from one sample to the other at the expense of some slower movement relative to 'simple' XY-movement. The TICs provide a general indication for the rapid acquisition now possible using this simple MAI-MS approach for synthetic polymer analyses using the 3-NBN matrix (**Figure 5B**; and **Figures S19B** and **S20B, supporting information**). Without optimization, the reproducibility is good (**Figure S19, supporting information**) and carryover is surprisingly low (**Figure S20, supporting information**) considering the past difficulties using the matrix 3-NBN for polymer analyses on this instrument which stresses the effectiveness of the retrofitted MSTM heated inlet tube.⁷⁴ This contactless 'vacuum cleaner' approach based on *inlet* ionization operates conveniently between 10 and 15 seconds per sample, shown in **Figure S19B (supporting information)**, compared with several minutes using *vacuum* ionization conditions, such as shown in **Figure 1B**. In

some cases, the addition of salt slows down the MAI operation to some extent (of the order of seconds), which may be related to the relatively larger matrix/analyte/salt crystals formed when metal salt is added, but usually provides the desired straightforward interpretation of the $[M+ncation]^{n+}$ as is shown in **Figure 5C**. The apparent Li cation addition to PEGDME-1500 (**Figure S21, supporting information**), readily calculates to the doubly- and singly-charged polymer ions with little to no chemical background. The concern remains that there is physical contact between the salt-containing matrix:analyte and the narrow inlet of the mass spectrometer so that one must expect similar challenges to those known with direct injection ESI.

These vMAI results from atmospheric and intermediate pressure are compelling evidence for eliminating the traditional 'ion source' of voltage and/or laser, reducing pumping capacity, and providing ease of operation. Carryover and cross contamination have not been fully addressed employing certain commercial ESI ion sources that tend to be especially problematic with direct injection methods as is the case using traditional ionization means. Therefore, the method developments described here for MAI (not vMAI) must be expected to be associated with cross-contamination and eventual source contamination, especially because of the usual necessity of the addition of an involatile salt for polymer characterization.^{11-22,24-26} Based on these proof-of-principle results using the new ionization processes here, it is unlikely that commercial ion sources built for ESI, APCI and MALDI operation will be ultimately successful in the broader applicability of (v)MAI, solvent assisted ionization, and laserspray ionization. Consequently, an immediate need to be addressed is the development of dedicated sources. Robustness associated with the new ionization methods is more likely to be found with *vacuum* ionization methods, which avoid contact with source elements, if simpler ways than a 'vacuum MALDI source without the use of a laser' can be developed.

Method Development using Dedicated Mass Spectrometer Sources

Commercial ion sources optimized for use of high voltages, lasers, and desolvation gases have been developed now for 30 years. Conversely, the new ionization processes that depend on temperature, pressure differentials and associated collisions²⁸⁻³² are more likely to be enhanced through the development of dedicated sources instead of 'reusing' or 'repurposing' traditional commercial ion sources. Here, we focus our attention to the characterization of polymers up to 2 kDa and plasticizers by exploiting the design and construction of dedicated sources serving the new ionization processes and the specific needs associated with different synthetic compound classes.

As is the case with any direct injection method in MS, especially when it includes a narrow channel (inlet tube), contamination from sample-to-sample is always an obstacle and this may increase significantly with salts added during sample preparation. This becomes more problematic the longer the inlet tube is, although increased heat can be applied to mitigate the issue. Furthermore, it can be expected that minimizing the sample consumption will reduce some of these challenges. A prototype vacuum MAI source based on a *vacuum*-probe introduction has been recently constructed so that the sample mixture does not touch physical source components.⁶⁹ Placing a drop of the matrix/analyte/salt solution on a probe tip and exposing it to sub-atmospheric pressure (typically 0.2-0.5 mbar) in the prototype *vacuum* source produces the typical ESI-like multiply-charged polymer ions. PEG-1000 and PEGDME-1500, two polymers differing by end group, were ionized with this method using different salts (Li^+ , Na^+ , Rb^+ , Cs^+) for cationization. As the size of these monovalent cations increased, the ability to attach to the polymer became increasingly difficult (**Figure S22, supporting information**) as expected.

The cation used to ionize polymers was further studied with the *vacuum*-probe because of the known critical role in enabling MS detection through cationization of

these basic non-volatile compounds.⁵⁸⁻⁶⁷ **Figure 6** compares the ionization rate of PEG-1000 and 1,2-DCB matrix with added barium chloride and sodium trifluoroacetate. With sodium trifluoroacetate salt, the TIC was observed to have reached a maximum of 2.03×10^4 counts after 7.5 minutes, and ceased after 12 minutes (**Figure 6A1**). Using barium chloride as the salt while keeping everything else constant, the TIC readout reached a maximum of 9.92×10^4 counts after only one minute, and decreased slowly over the course of 23 minutes (**Figure 6A2**). With this ion abundance it is possible to obtain a good spectrum from a single, 0.5 s scan at the apex of the TIC (**Figure S23, supporting information**). The barium-induced cationization of PEG was confirmed by the unique isotopic pattern (**Figure 6C2**). The benefits of barium cationization were also observed using the matrix methyl 2-methyl-3-nitrobenzoate by comparing PEG-1000 without salt (**Figure S24I, supporting information**), with lithium chloride (**Figure S24II, supporting information**), and with barium chloride (**Figure S24III, supporting information**). The ion abundance was observed to be an order of magnitude higher in the barium-cationized PEG than in either of the other matrix/analyte(/salt) combinations. In addition to the high ion abundance, using the same matrix the sample containing the salt additions revealed a slightly more rapid onset of polymer ionization. In case of the barium addition (**Figure S24IIIA, supporting information**), the polymer ionization continued for nearly twice as long as with the other conditions used here (**Figure S24I and IIA, supporting information**). Barium cationization favored primarily even numbered charge states, presumably because of its divalency (**Figure S24IIIB1, supporting information**). Trials performed with barium salts in both methyl 2-methyl-3-nitrobenzoate and 1,2-DCB matrices (**Figure 6**) therefore provided TICs with more favorable characteristics than with monovalent salts: barium adduction (**Figure 6IIB1; and Figure S24III, supporting information**) commences ion formation more quickly, reaches a higher maximum ion abundance, and continues to ionize over a longer time span than sodium- (**Figure 6I**) or lithium-adducted (**Figure S24II, supporting**

information) low molecular weight PEG. While divalent cations such as strontium (**Figure S25, supporting information**) and in particularly barium tend to enable more rapid, more abundant, and longer ionization of the polymer PEG-1000 than monovalent cations such as sodium or lithium, the cause is unknown at this time. Divalent cations have been seldom used in MS.⁶⁵

We hypothesize that the sustained polymer ion abundance, as is also the case with ESI, is of excellent analytical value, especially when samples with ever-increasing complexity are encountered. The continuous ion formation of vMAI is reminiscent of that of ESI but from the solid state and without the disadvantages associated with direct injection ESI of complex, 'dirty', and/or salty samples such as capillary clogging, contamination, and carryover issues. Contrary to ESI and the use of the new ionization processes on commercial ESI source inlets (*inlet* ionization), these direct injection conditions of a *vacuum*-probe device based on vMAI minimize or eliminate the potential for cross- or source contamination. Re-analyses are possible, simply by adding more matrix to the probe before the probe tip is cleaned off. For a new sample, either the probe tip needs to be cleaned, often as simple as rinsing it with copious amounts of deionized water for ca 10 s, or the probe replaced. Using the *vacuum*-probe, the matrix/analyte/salt introduction is prompt (few seconds), contrary to commercial vacuum MALDI sources which require at least 2 minutes for the vacuum to pump down the plate region. However, depending on the source configuration of the commercial mass spectrometer, the onset of ion formation can be slow, as already seen above with the addition (or omission) of different salts and/or matrices. The ion formation can be manipulated to some degree in terms of ion abundance and duration by changing the gas flow thereby increasing or decreasing the collisions of the charged clusters with gases. However, another effective means is the use of voltages as was demonstrated on a vacuum MAI source based on a *vacuum*-probe retrofitted on a Thermo mass spectrometer.⁷⁰ Using this recently constructed prototype *vacuum*-probe MAI source on

the Q-Exactive Focus, fast and accurate mass measurements are possible, such as the determination of lithium adduction in PEG-1500 (**Figure 7A**), or the confirmation of ammonium and sodium adducted bisphenol A species (**Figure 7B**) which indicates that ionization of this plasticizer is readily achieved. With and without the addition of salts, ammonium and sodium adducted plasticizer ions were observed. A packing material was briefly extracted with acetone and ionized using the *vacuum*-probe approach (**Figure 7C**). While the matrix related signals appear more pronounced on this instrument than on the Waters instrument using the respective *vacuum*-probe sources, the ultra-high resolution readily identifies the phthalic acid dibutyl ester, erucamide, phthalic acid dioctyl ester, and potentially oxidized Irgafos 168 (**Figure 7C**) without the need for external standards such as those used with the Waters SYNAPT G2S (**Figures S9 and S10, supporting information**).

Vacuum-probe (*vacuum* ionization) is more universally applicable for polymer characterization than *inlet* ionization approaches as was shown here for two fundamentally different mass spectrometers (Waters Q-IMS-TOF, Thermo Orbitrap). As noted above, one of the major concerns in analyzing polymers using MS is the potential of instrument contamination. To specifically investigate the benefits of using the simplified *vacuum*-probe source, for sample preparation, a premixed matrix/analyte was simply dropped onto the probe tip indicating that the sample preparation is rather forgiving. To overcome the potential issue and concern of carryover and instrument conditions, fexofenadine was analyzed by simply cleaning the probe tip immediately after the analysis of polymers/plasticizers using the *vacuum*-probe on the Orbitrap Q-Exactive Focus. As observed in **Figure S26 (supporting information)**, no carryover signals related to polymers or plasticizers were detected, indicating that the analysis performed using the simplified *vacuum* ionization source is free from instrument contamination, independent of the sample type. It also demonstrates the principal

applicability of vMAI for polymer characterization on other instruments and offers wider developments in the future.

In a final experiment, a tertiary blend of low-molecular weight PEG, plasticizer, and property modifier typically present to stabilize materials was examined using 3-NBN matrix and NaTFA salt with the prototype *vacuum*-probe vMAI source on a Waters SYNAPT G2S IMS-TOF instrument. In **Figure 8** doubly-charged PEG-1000 ions can be observed that are separated in the gas phase from plasticizer PEG-EH and the property modifier DADBU as is readily visualized by the 2-D plot of drift time versus m/z , or in a more in-depth analyses by the extracted mass spectra (**Figure S27, supporting information**). The success of the IMS-MS gas-phase separation approach is related to the formation of the multiply-charged polymer ions formed by vMAI, similar to previous work using ESI-IMS-MS.^{14,15} The results are comparable with those from a heated *inlet* ionization approach using MAI-IMS-MS.⁴⁴ However, vMAI-IMS-MS provides added convenience and exceptional confidence in robust measurements even of 'dirty' and high salt content samples, contrary to concerns when using traditional and *inlet* ionization approaches. One can therefore expect product control to be sufficient using a (ultra)high resolution vMAI-IMS-MS approach directly from its native environment such as a polymer film from a surface or solid state, without added energy or the inconvenience of the commercial ESI and MALDI sources, to give the highly desirable gaseous ions of polymers and polymer additives, respectively, that separate efficiently in the gas phase by IMS-MS because of the unique ability to form the multiply charged-ions directly from a surface.

CONCLUSION

A key issue for polymer characterization using an MS approach is the efficient conversion of these frequently neutral and less basic (macro)molecules to gas-phase ions which can be achieved through typically monovalent metal salt additives.^{2,3,58-61,63,65-68} We provided a progress report on new ways to analyze synthetic polymers and related small molecule additives simply using a matrix and a salt additive, preferably divalent. Current challenges in commercial sources relative to dedicated sources and their applicability to a variety of mass spectrometers simply using the pressure (differential) and/or temperature are exemplified, providing exceptionally simple means for analyses. Common to these new ionization methods is that the traditional 'ion source' is eliminated and that the difficulties of applying the new processes to polymer analyses,⁴⁴ restrictions associated with atmospheric pressure to vacuum transfer to form gaseous ions, are overcome by making more effective use of the vacuum. However, the concern of contamination using *inlet* ionization remains as is the case with any direct injection approach where surface contact is a possibility. Both *vacuum* ionization, standing out for effectiveness and robustness, and *inlet* ionization, for easy access, were utilized and it is anticipated that further source developments will be able to address and combine both *vacuum* and *inlet* ionization advantages in one simple source. We also anticipate, based on previous work with biological polymers,^{38,53} analytical improvements through use of other MAI matrix(es), solvent(s), and salt combination(s). This contribution we view as a progress towards effective use of the new ionization processes. The currently developed source and methods provide instantaneous ionization, even using divalent cations, additional gas-phase separation revealing a unique 2-D pictorial snapshot using IMS-MS, just like ESI methods for polymer ionization¹⁵ and similar to MALDI but without a laser, that may reflect materials composition of as little as architectural differences of synthetic polymers. To move forward to higher dynamic range and wider applicability and finally to a state of high

spatial and temporal resolution and mass spectrometer portability to help meet the next generation materials development goals, improved fundamental research needs to continue on understanding how ions are formed using unprecedented ionization processes, how the throughput of the methods can be further increased and the footprint continued to be lowered for synthetic polymer characterization for which relative little attention is provided to improve analytical methodology despite their importance to society.

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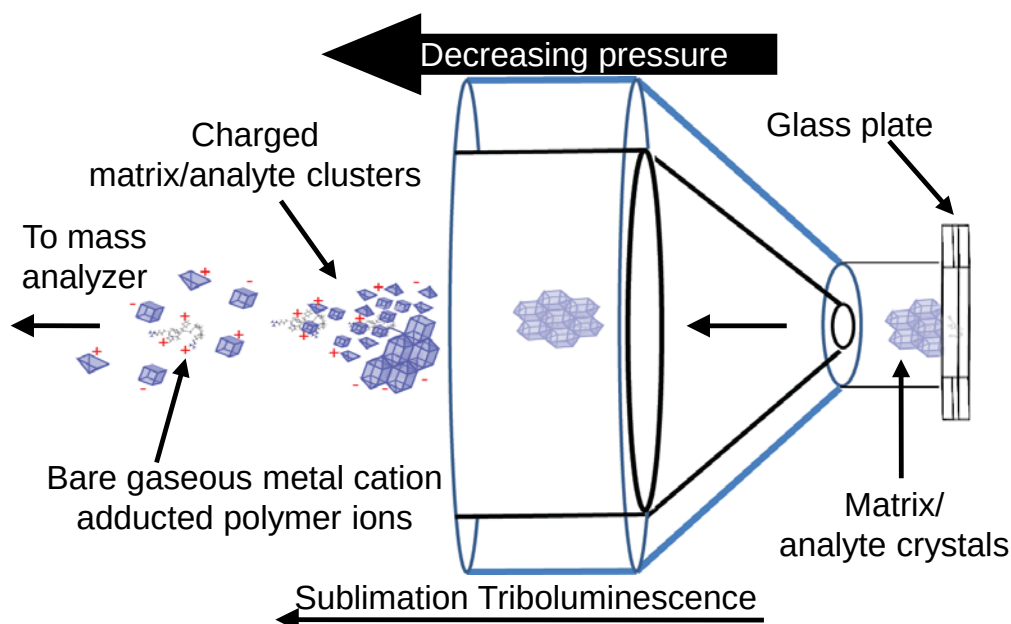
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Scheme 1: Schematic of the processes associated with MAI, here shown using the glass plate introduction method directly from atmospheric pressure using a modified skimmer inlet.



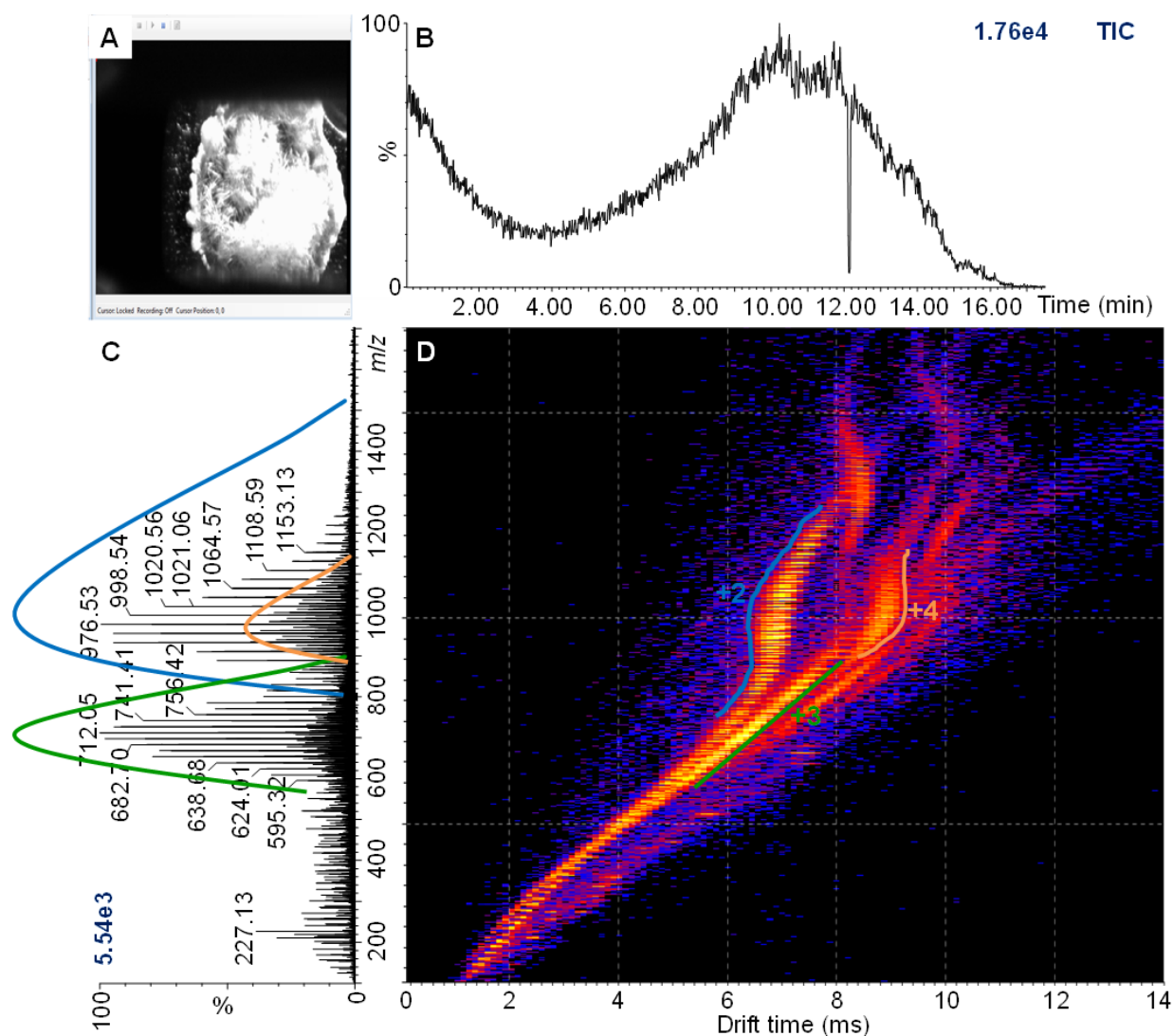


Figure 1. MALDI-IMS-MS of PEG-DME 2000 doped with LiTFA (9:1 v/v ratio) with 3-NBN matrix: (A) screenshot of the crystallized PEG-DME 2000:LiTFA (1 μ L) and 3-NBN matrix (1 μ L) on the sample metal plate, (B) total ion chromatogram, (C) total mass spectrum and (D) 2-dimensional plot of drift time versus m/z showing the +4, +3 and +2 charge states. Data acquired using MALDI settings on the intermediate pressure MALDI source of Waters SYNAPT G2S mass spectrometer. For more details, refer to **Figure S6**.

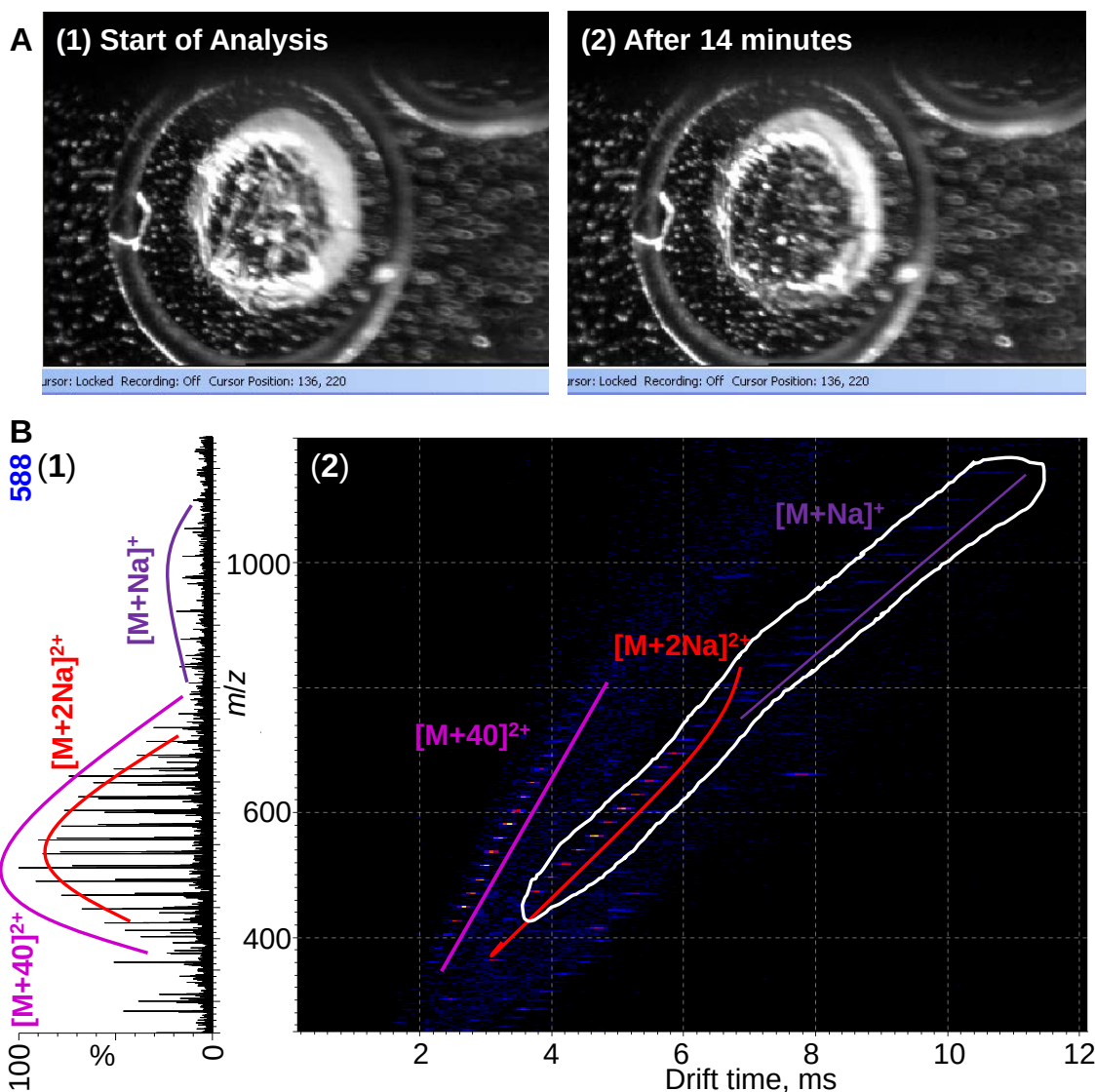


Figure 2. MAI-IMS-MS of linear PEG with bronopol matrix using an intermediate pressure MALDI source without the use of a laser: (A) screenshot commercial sample plate in the vacuum region of the source used to monitor the spontaneous ionization process without the addition of energy (1) at the beginning and (2) end of acquisition, (B) ions detected and displayed by (1) total mass spectrum and (2) IMS-MS 2-dimensional plot of drift time *versus* m/z . For more details, refer to **Figure S7** and **S8**.

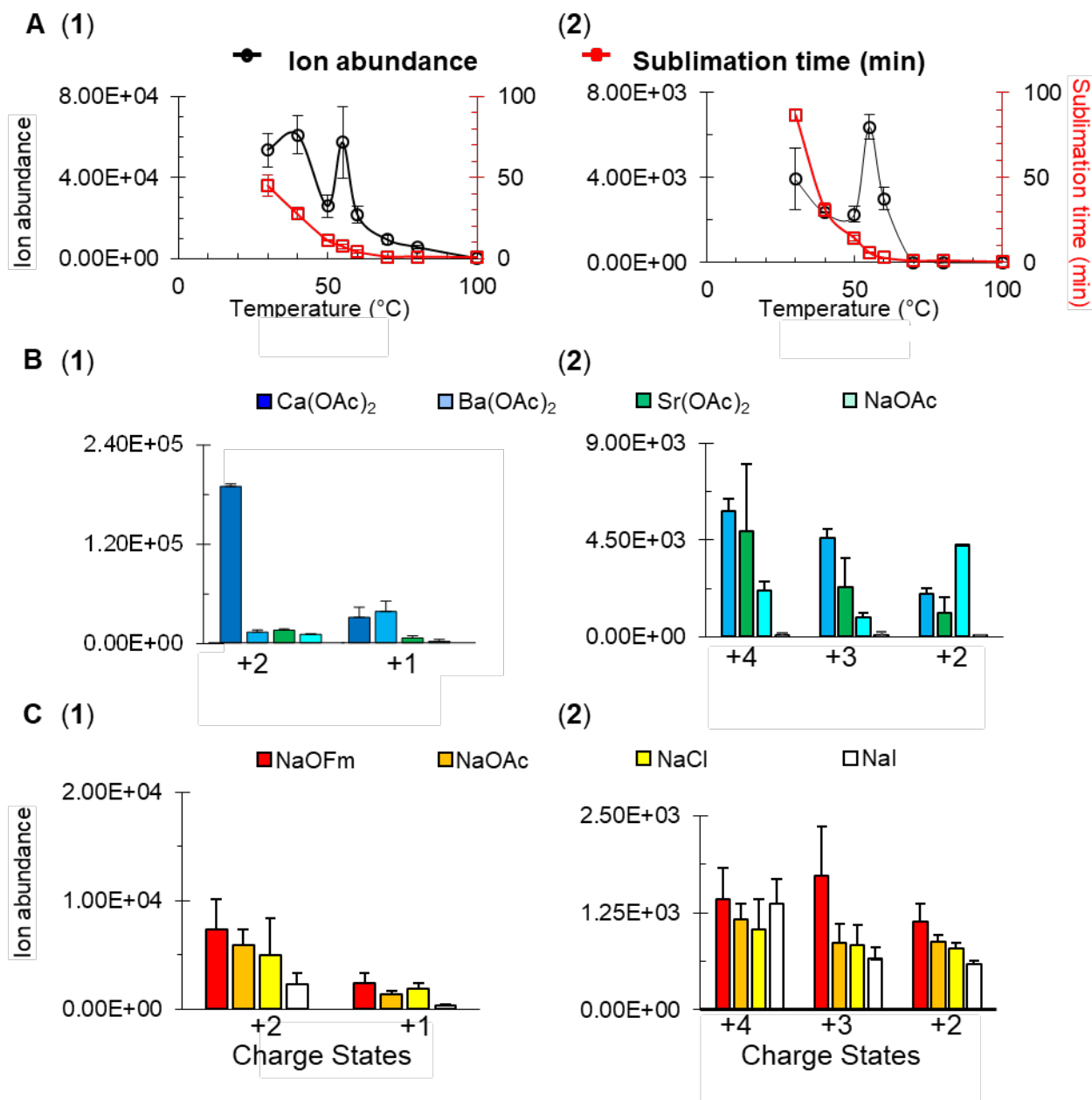


Figure 3. Ion abundance and sublimation time using the MAI matrix 2-methyl-2-nitro-1,3-propanediol and the role of (A) temperature using Ba(OAc)₂, (B) cation acetates, and (C) sodium counter anions, respectively, used for ionization of (1) PEG-1000 and (2) PEG-3000 on a Waters SYNAPT G2 mass spectrometer (Scheme S1B, Movie S1) using a source block temperature of 55 °C. For more details, refer to Figure S13.

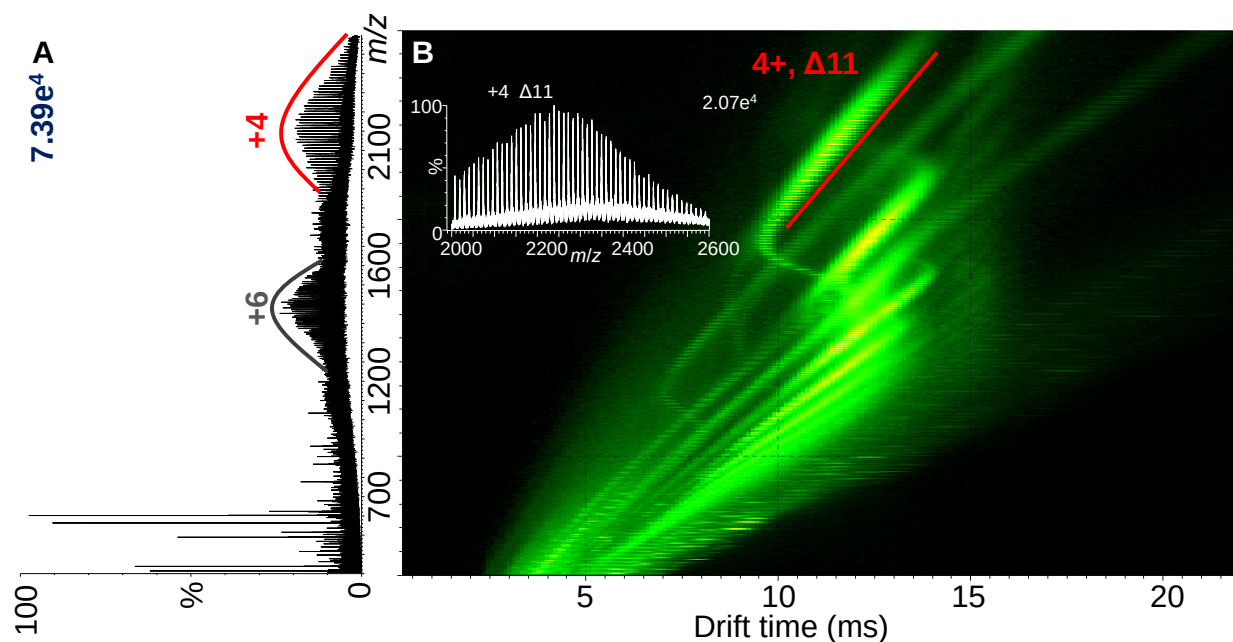


Figure 4. vMAI-IMS-MS of PEG-10000 and calcium II acetate salt in a 1:100 molar ratio and 2-methyl-2-nitropropane-1,3-diol matrix: (A) total mass spectrum and (B) 2-D plot display of drift time *versus* m/z with inset of the extracted +4 charge state. Data acquired on a Waters SYNAPT G2 mass spectrometer (**Scheme S1B**, **Movie S1**). For more details, refer to **Figure S14**.

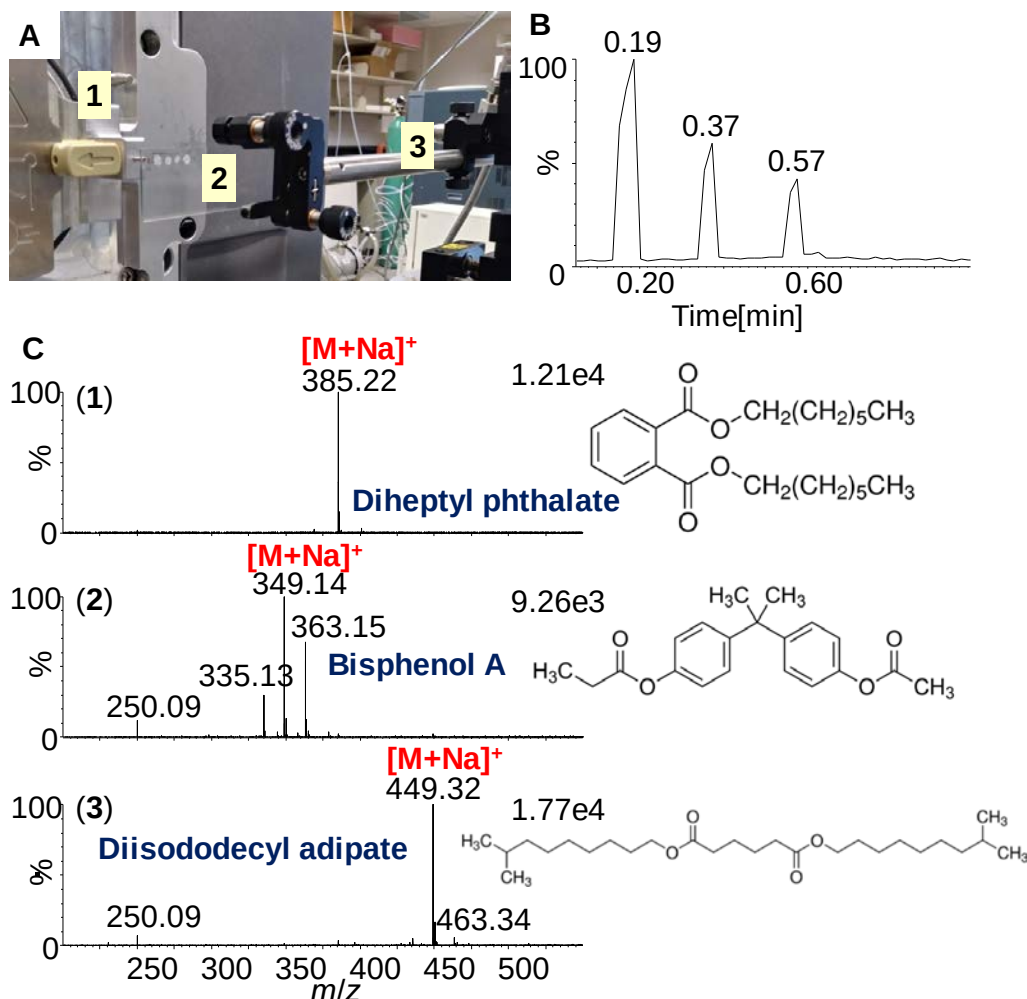


Figure 5. Automated, contactless MAL operation directly from atmospheric pressure: **(A)** photograph of the setup **(1)** MSTM heated inlet tube, **(2)** crystallized sample spots on glass plate and **(C)** XYZ-stage for automation; **(B)** total ion chromatogram and **(C)** mass spectra of **(1)** diheptyl phthalate, **(2)** bisphenol A and **(3)** diisododecyl adipate acquired consecutively. Data acquired with the matrix 3-NBN on a Waters SYNAPT G2S mass spectrometer. The analytes were mixed with 0.5 μL of 1 mg mL^{-1} NaCl solution prior to addition of 3-NBN matrix to facilitate sodium adduct formation. For more details, refer to **Figure S19-21**.

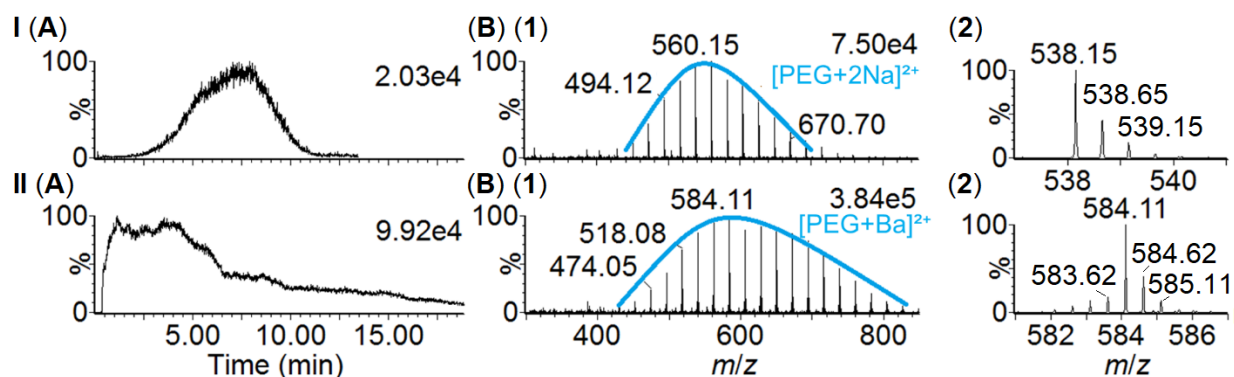


Figure 6. vMAI-MS analyses of PEG-1000 utilizing (I) NaTFA and (II) BaCl₂ salts for cationization. (A) TIC, (B) (1) mass spectrum of the PEG-1000 distribution, with (2) extracted isotopic distributions of the [HO(CH₂CH₂O)₂₃H+*n*Metal]²⁺ ions. Data acquired with the matrix 1,2-DCB on the dedicated *vacuum-probe*⁷⁰ fitted to an uncalibrated Waters SYNAPT G2S mass spectrometer. Refer to **Figure S23** for more details.

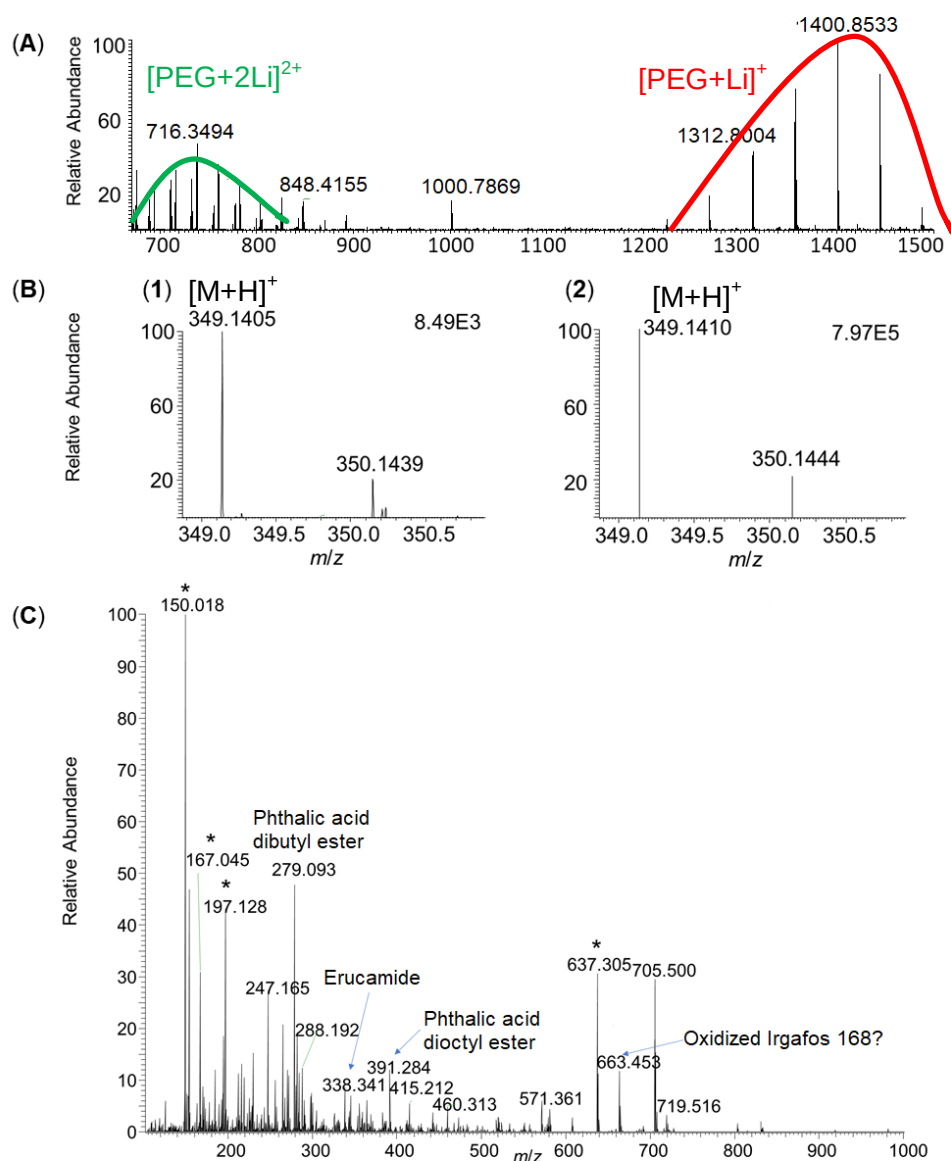


Figure 7. Accurate mass measurements on an uncalibrated mass spectrometer using the matrix 3-NBN: (A) PEG-1500 ionized in the presence of lithium chloride, (B) plasticizer bisphenol A (1) experimental and (2) modeled isotopic distribution, and (C) packing material extracted with acetone. Asterisks mark peaks associated with the matrix 3-NBN. Data acquired using the dedicated *vacuum* source on a Thermo Q-Exactive Focus.

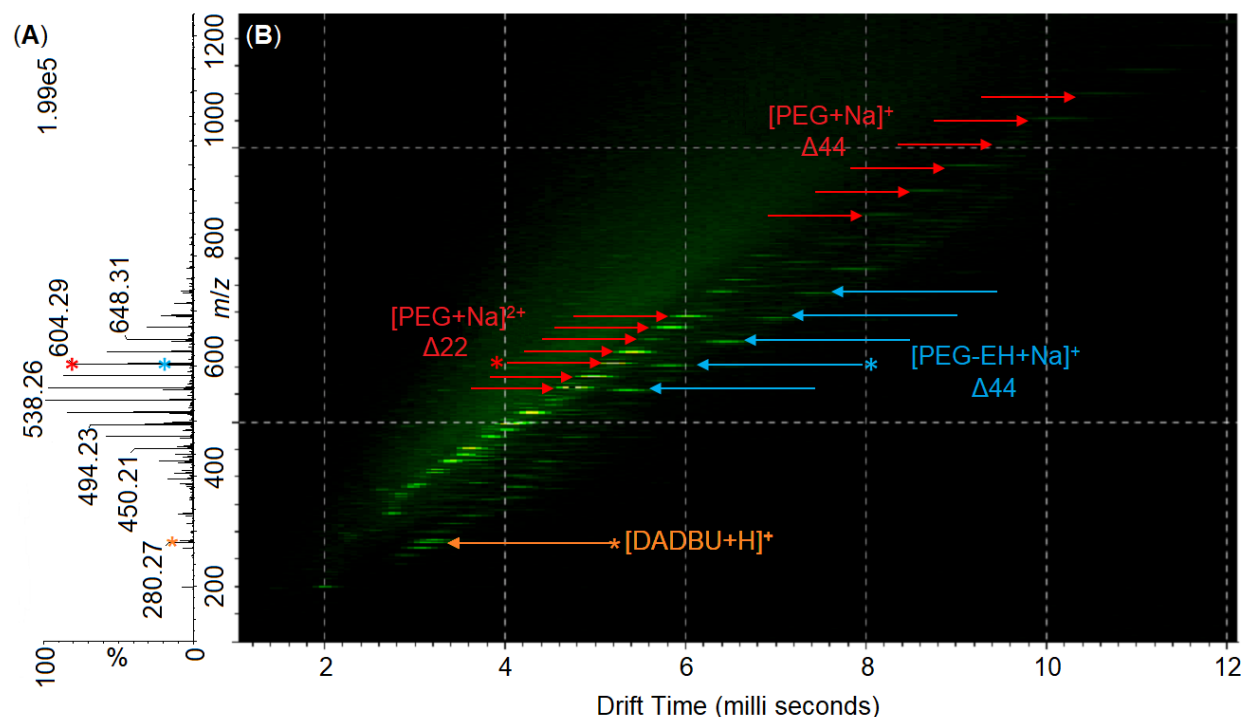


Figure 8. vMAI-IMS-MS approach to the analysis of a tertiary mixture containing a polymer, plasticizer, and property modifier. **(A)** total mass spectrum and **(B)** 2-D plot of drift time *versus* m/z containing PEG-1000 (red), plasticizer PEG-EH 640 (blue), and property modifier DADBU (yellow). Data acquired with vMAI using 3-NBN matrix and NaTFA salt on the *vacuum-probe*⁷⁰ fitted on a Waters SYNAPT G2S. For reference, colored asterisks match signal in 2D plot of drift time *versus* m/z with the corresponding peak in the mass spectrum. For more details, refer to **Figure S27**.